

International Cybersecurity and Digital Forensics Academy (ICDFA)

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2025/FWSD/11478

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SBT-DF203-Lab4_2025/FWSD/11478_Husseini Suleiman

SBT-DF203: Basic Networking Skills for Digital Forensics

LAB 4: SMTP Email Traffic Forensics

Three-Week Delivery Block 2/3 | 12-18 September 2026

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Important Pre-Lab Instruction Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.	
Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 4
Lab Title	SMTP Email Traffic Forensics
Delivery Block	2/3 of 3
Scheduled Dates	12-18 September 2026
Estimated Duration	3-4 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	smtp.pcap supplied sample capture

1. Source Materials and Slide Alignment

- Email Traffic Forensics: SMTP purpose, ports 25/587/465, plaintext commands, STARTTLS/SMTPS distinctions, Base64 encoding, stream reassembly and email-content analysis.
- The slide practical uses smtp.pcap containing Microsoft Outlook traffic and requires timestamps, client, content, ports, IP addresses and MAC addresses.
- The exercise treats credentials as sensitive evidence and requires protected reporting.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

A historical SMTP packet capture has been supplied for analysis. You must determine when the email exchange occurred, identify the client and server, reconstruct SMTP commands and response codes, decode relevant Base64 authentication fields, recover the message content, and document the network path without exposing sensitive data unnecessarily.

3. Learning Outcomes

- Explain the role of SMTP and distinguish plaintext SMTP, STARTTLS and implicit TLS.
- Identify SMTP commands, server response codes and authentication exchanges.
- Reassemble an SMTP TCP stream and reconstruct message headers and body.
- Decode Base64 values in a controlled offline manner.
- Extract timestamps, client software, IP addresses, ports and MAC addresses.
- Assess evidential limitations when encryption is used or capture is incomplete.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.
- Treat any recovered username, password or message content as confidential assessment evidence. Mask sensitive values in screenshots unless the instructor explicitly requires the full value.

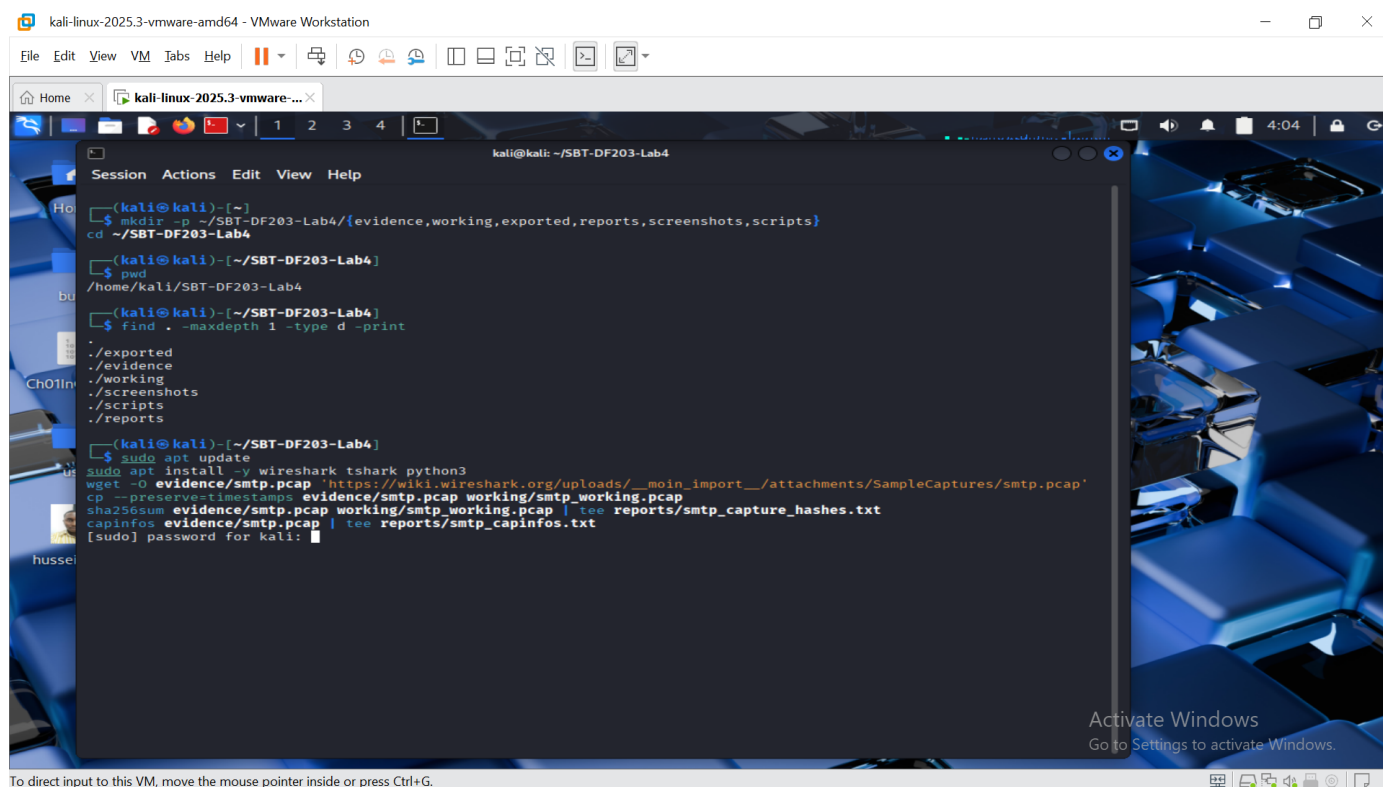
5. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux or Ubuntu VM	Analyze the capture.
Packet tools	Wireshark and tshark	Filter and reconstruct SMTP streams.
Decoder	Python 3 base64 module	Decode Base64 strings offline.
Hashing	sha256sum	Preserve evidence integrity.
Evidence	smtp.pcap	Historical training traffic.
Text editor	Any local editor	Document recovered message safely.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab4/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab4
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.



```

sudo apt update
sudo apt install -y wireshark tshark python3
wget -O evidence/smtp.pcap
'https://wiki.wireshark.org/uploads/___moin_import___/attachments/SampleCaptures/smtp.pcap'
cp --preserve=timestamps evidence/smtp.pcap working/smtp_working.pcap
sha256sum evidence/smtp.pcap working/smtp_working.pcap | tee reports/smtp_capture_hashes.txt
capinfos evidence/smtp.pcap | tee reports/smtp_capinfos.txt

```

Screenshot required: smtp.pcap file details, capinfos summary and matching original/working-copy hashes.

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab4-Husseini Suleiman
Trainee name	Husseini Suleiman
Date and time started	2026-09-14 04:09:12 (978 KB/s) - 'evidence/smtp.pcap' saved [27850/27850]
Evidence file name(s)	SBT-DF203-Lab4
Source or generation method	Kali command line
Original SHA-256	17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
Working-copy SHA-256	3def1a1fed849e7f23b66e6925ddff05845a400b
Analysis workstation/VM	kali@kali)-[~/SBT-DF203-Lab4]
Notes on any changes	nil

8. Part A - Inventory the Capture and Locate SMTP Streams `tshark -r working/smtp_working.pcap -q -z conv,tcp | tee reports/tcp_conversations.txt`

```
tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
-e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e _ws.col.Info \
| tee reports/smtp_packet_inventory.tsv
```

Record of the following

The SMTP server port: 25

Client ephemeral port: 465

First packet time: DATA fragment, 1460 bytes

Last packet time: 245 bytes

Number of SMTP frames: 53 24KB

The screenshot shows a Kali Linux terminal window with the following commands and output:

```
(kali@kali)~[/SBT-DF203-Lab4]
$ tshark -r working/smtp_working.pcap -q -z conv,tcp | tee reports/tcp_conversations.txt
tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
-e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e _ws.col.Info \
| tee reports/smtp_packet_inventory.tsv
Warning: program compiled against libxml 215 using older 214
```

The output shows a table of TCP Conversations. The first entry is between 10.10.1.4:1470 and 74.53.140.153:25, showing 25 frames and 1,980 bytes. The second entry is between 10.10.1.4:1470 and 74.53.140.153:25, showing 28 frames and 22 kB. The third entry is between 10.10.1.4:1470 and 74.53.140.153:25, showing 53 frames and 24 kB. The output also includes a list of SMTP messages, including an EHLO message, an AUTH LOGIN message, and a DATA message.

Filter: <No Filter>	Frames	Bytes	Frames	Bytes	Total	Relative	Duration
	Frames	Bytes	Frames	Bytes	Frames	Start	
10.10.1.4:1470 ↔ 74.53.140.153:25	25	1,980 bytes	28	22 kB	53	0.036986000	7.5777

Warning: program compiled against libxml 215 using older 214

Size	Time	Source	Destination	Protocol	Length	Info
6	2009-10-05T02:06:08.219663000-0400	74.53.140.153	10.10.1.4	25	1470	S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 We do not authorize the use of this system to transport unsolicited, and/or bulk e-mail.
7	2009-10-05T02:06:08.224899000-0400	10.10.1.4	74.53.140.153	25	1470	C: EHLO GP
9	2009-10-05T02:06:08.566183000-0400	74.53.140.153	10.10.1.4	25	1470	S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157]
10	2009-10-05T02:06:08.568729000-0400	10.10.1.4	74.53.140.153	25	1470	C: AUTH LOGIN
11	2009-10-05T02:06:08.911081000-0400	74.53.140.153	10.10.1.4	25	1470	S: 334 VXNlcm5hbWU6
12	2009-10-05T02:06:08.911655000-0400	10.10.1.4	74.53.140.153	25	1470	C: User: Z3VycGFydGFwQHdHJpb3RzLmlu
13	2009-10-05T02:06:09.253544000-0400	74.53.140.153	10.10.1.4	25	1470	S: 334 UGFzc3dvcmQ6
14	2009-10-05T02:06:09.254118000-0400	10.10.1.4	74.53.140.153	25	1470	C: Pass: cHVuamFiQDEyMw==
15	2009-10-05T02:06:09.613798000-0400	74.53.140.153	10.10.1.4	25	1470	S: 235 Authentication succeeded
16	2009-10-05T02:06:09.614414000-0400	10.10.1.4	74.53.140.153	25	1470	C: MAIL FROM: <gurpartap@patriots.in>
17	2009-10-05T02:06:09.956765000-0400	74.53.140.153	10.10.1.4	25	1470	S: 250 OK
18	2009-10-05T02:06:09.957250000-0400	10.10.1.4	74.53.140.153	25	1470	C: RCPT TO: <raj_deol2002in@yahoo.co.in>
19	2009-10-05T02:06:10.319708000-0400	74.53.140.153	10.10.1.4	25	1470	S: 250 Accepted
20	2009-10-05T02:06:10.320203000-0400	10.10.1.4	74.53.140.153	25	1470	C: DATA
21	2009-10-05T02:06:10.661679000-0400	74.53.140.153	10.10.1.4	25	1470	S: 354 Enter message, ending with "." on a line by itself
22	2009-10-05T02:06:10.692743000-0400	10.10.1.4	74.53.140.153	25	1470	C: DATA fragment, 1460 bytes
23	2009-10-05T02:06:10.692786000-0400	10.10.1.4	74.53.140.153	25	1470	C: DATA fragment, 1460 bytes
24	2009-10-05T02:06:10.692804000-0400	10.10.1.4	74.53.140.153	25	1470	C: DATA fragment, 1460 bytes
25	2009-10-05T02:06:10.692823000-0400	10.10.1.4	74.53.140.153	25	1470	[TCP Window Full] C: DATA fragment, 1460 bytes
26	2009-10-05T02:06:10.695115000-0400	192.168.1.1,10.10.1.4	10.10.1.4,74.53.140.153	25	25	Destination unreachable (Fragmentation needed)
28	2009-10-05T02:06:10.695623000-0400	192.168.1.1,10.10.1.4	10.10.1.4,74.53.140.153	25	25	Destination unreachable (Fragmentation needed)
29	2009-10-05T02:06:10.696248000-0400	192.168.1.1,10.10.1.4	10.10.1.4,74.53.140.153	25	25	Destination unreachable (Fragmentation needed)

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```

kali@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help
9 2009-10-05T02:06:08.566183000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] |
SIZE 52428800 | PIPELINING | AUTH PLAIN LOGIN | STARTTLS | HELP
10 2009-10-05T02:06:08.568729000-0400 10.10.1.4 1470 74.53.140.153 25 C: AUTH LOGIN
11 2009-10-05T02:06:08.911081000-0400 74.53.140.153 25 10.10.1.4 1470 S: 334 VXNlcm5hbWU6
12 2009-10-05T02:06:08.911655000-0400 10.10.1.4 1470 74.53.140.153 25 C: User: Z3VycGFydGFWQHBhdHJpb3RzLmlu
13 2009-10-05T02:06:09.253544000-0400 74.53.140.153 25 10.10.1.4 1470 S: 334 UGFzc3dvcmQ6
14 2009-10-05T02:06:09.254118000-0400 10.10.1.4 1470 74.53.140.153 25 C: Pass: cHVuamFiQDEyMw==
15 2009-10-05T02:06:09.613798000-0400 74.53.140.153 25 10.10.1.4 1470 S: 235 Authentication succeeded
16 2009-10-05T02:06:09.614414000-0400 10.10.1.4 1470 74.53.140.153 25 C: MAIL FROM: <gurpartap@patriots.in>
17 2009-10-05T02:06:09.958765000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250 OK
18 2009-10-05T02:06:09.957250000-0400 10.10.1.4 1470 74.53.140.153 25 C: RCPT TO: <raj_deol2002in@yahoo.co.in>
19 2009-10-05T02:06:10.319708000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250 Accepted
20 2009-10-05T02:06:10.320203000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA
21 2009-10-05T02:06:10.661679000-0400 74.53.140.153 25 10.10.1.4 1470 S: 354 Enter message, ending with "." on a line by itself
22 2009-10-05T02:06:10.692743000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1460 bytes
23 2009-10-05T02:06:10.692786000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1460 bytes
24 2009-10-05T02:06:10.692804000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1460 bytes
25 2009-10-05T02:06:10.692823000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Window Full] C: DATA fragment, 1460 bytes
26 2009-10-05T02:06:10.695115000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragmentation needed)
27 2009-10-05T02:06:10.695623000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragmentation needed)
28 2009-10-05T02:06:10.696248000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragmentation needed)
29 2009-10-05T02:06:10.696634000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragmentation needed)
30 2009-10-05T02:06:11.104972000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
31 2009-10-05T02:06:11.104998000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
32 2009-10-05T02:06:11.469759000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
33 2009-10-05T02:06:11.469814000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
34 2009-10-05T02:06:11.494181000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
35 2009-10-05T02:06:11.494199000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
36 2009-10-05T02:06:11.834628000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
37 2009-10-05T02:06:11.834655000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
38 2009-10-05T02:06:11.858316000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
39 2009-10-05T02:06:11.858334000-0400 10.10.1.4 1470 74.53.140.153 25 from: "Gurpartap Singh" <gurpartap@patriots.in>, subject: SM
TP. (text/plain) (text/html) (text/plain) (text/plain) (text/plain) |
52 2009-10-05T02:06:12.248789000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250 OK id=1Mugho-0003Dg-Un
53 2009-10-05T02:06:14.763825000-0400 10.10.1.4 1470 74.53.140.153 25 C: QUIT
54 2009-10-05T02:06:15.105467000-0400 74.53.140.153 25 10.10.1.4 1470 S: 221 xc90.websitewelcome.com closing connection
(kali@kali)~$

```

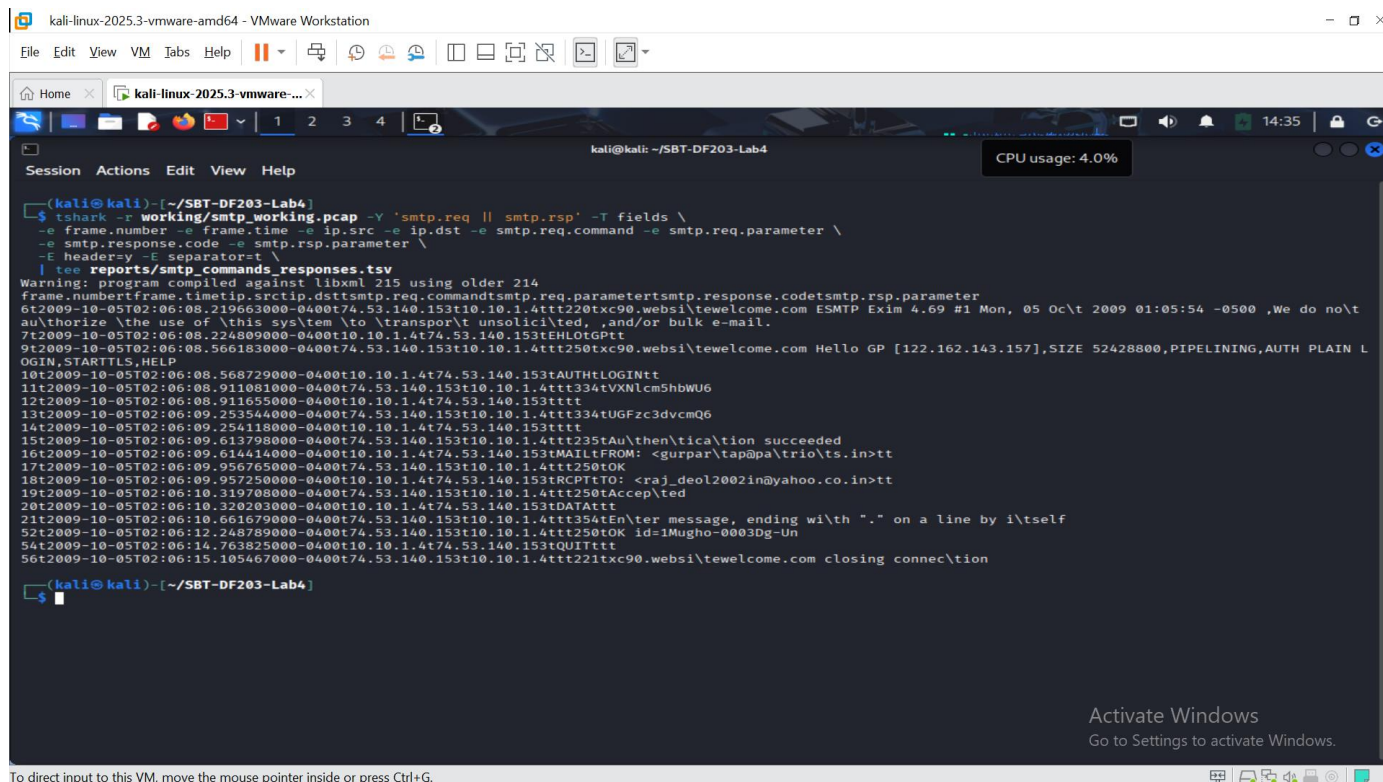
cHVuamFiQDEyMw==

9. Part B - Identify Commands and Response Codes

Sequence	Direction	Command/Code	Meaning	Timestamp
1	Server to client	220	Service ready	7t2009-10-05T02:06:08.224809000-0400t10.10.1.4t74.53.140.153tEHLOtGPtt
2	Client to server	EHLO/HELO	Client greeting	7t2009-10-05T02:06:08.224809000-0400t10.10.1.4t74.53.140.153tEHLOtGPtt
3	Client to server	AUTH	Authentication method	10t2009-10-05T02:06:08.568729000-0400t10.10.1.4t74.53.140.153tAUTHtLOGIntt
4	Client to server	MAIL FROM	Envelope sender	16t2009-10-05T02:06:09.614414000-0400t10.10.1.4t74.53.140.153tMAILtFROM:<gurpartap@patriots.in>tt
5	Client to server	RCPT TO	Envelope recipient	18t2009-10-05T02:06:09.957250000-0400t10.10.1.4t74.53.140.153tRCPTtTO:<raj_deol2002in@yahoo.co.in>tt
6	Client to server	DATA	Message content begins	20t2009-10-05T02:06:10.320203000-0400t10.10.1.4t74.53.140.153tDATAttt
7	Client to server	QUIT	Session termination	54t2009-10-05T02:06:14.763825000-0400t10.10.1.4t74.53.140.153tQUITttt

```
tshark -r working/smtp_working.pcap -Y 'smtp.req || smtp.rsp' -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e smtp.req.command -e smtp.req.parameter \
-e smtp.response.code -e smtp.response.parameter \
| tee reports/smtp_commands_responses.tsv
```

```
tshark -r working/smtp_working.pcap -Y 'smtp.req || smtp.rsp' -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e smtp.req.command -e smtp.req.parameter \
-e smtp.response.code -e smtp.rsp.parameter \
-E header=y -E separator=t \
| tee reports/smtp_commands_responses.tsv
```



10. Part C - Decode Base64 Authentication Evidence

Base64 is encoding, not encryption. Copy only the required encoded values into the offline script. Do not use online decoders.

```
python3 - <<'PYLAB'
import base64
samples = {
    'username': 'REPLACE_WITH_CAPTURED_BASE64',
    'password': 'REPLACE_WITH_CAPTURED_BASE64'
}
for label, value in samples.items():
    try:
        decoded = base64.b64decode(value).decode('utf-8', errors='replace')
        print(f'{label}: {decoded}')
    except Exception as exc:
        print(f'{label}: decode failed: {exc}')
PYLAB
```

Reporting Rule

In the public-facing report body, show a masked value such as analyst@example.com and P*****3. Place full decoded values only in the protected evidence appendix if required by the instructor.

```

kali@kali: ~/SBT-DF203-Lab4
$ python3 - <<'PYLAB'
import base64
samples = {
    'username': 'Z3VycGFydGFwQHBhdHJpb3RzLmlu',
    'password': 'cHVuamFiQDEyMw=='
}
for label, value in samples.items():
    try:
        decoded = base64.b64decode(value).decode('utf-8', errors='replace')
        print(f'{label}: {decoded}')
    except Exception as exc:
        print(f'{label}: decode failed: {exc}')
PYLAB
username: gurpartap@patriots.in
password: punjab@123
kali@kali: ~/SBT-DF203-Lab4
$
  
```

Answer: Command used:

```

python3 - <<'PYLAB'
import base64
samples = {
    'username': 'Z3VycGFydGFwQHBhdHJpb3RzLmlu',
    'password': 'cHVuamFiQDEyMw=='
}
for label, value in samples.items():
    try:
        decoded = base64.b64decode(value).decode('utf-8', errors='replace')
        print(f'{label}: {decoded}')
    except Exception as exc:
        print(f'{label}: decode failed: {exc}')
PYLAB
  
```

HASH PASSWORD

```

username: gurpartap@patriots.in
password: punjab@123
  
```

11. Part D - Reconstruct the Email Message

1. In Wireshark select an SMTP packet and choose Follow > TCP Stream.

2. Identify SMTP envelope fields separately from RFC message headers.
3. Record Date, From, To, Subject, Message-ID, MIME-Version, Content-Type and User-Agent/X-Mailer where present.
4. Determine whether the body is plain text, HTML, multipart or contains an attachment.
5. Save a redacted text reconstruction to reports/reconstructed_email_redacted.txt.

Answer: SREAM number =0

```

kali@kali: ~/SBT-DF203-Lab4
$ tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,0 \
tee reports/smtp_stream_0.txt
Warning: program compiled against libxml 215 using older 214

Follow: tcp,ascii
Filter: tcp.stream eq 0
Node 0: 10.10.1.4:1470
Node 1: 74.53.140.153:25
181
220 xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220 We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.
9
EHLO GP
137
250 xc90.websitewelcome.com Hello GP [122.162.143.157]
250 SIZE 52428800
250 PIPELINING
250 AUTH PLAIN LOGIN
250 STARTTLS
250 HELP
12
AUTH LOGIN
18
334 VXNlcm5hbWU6
30
Z3VycGFydGwQHBhdHJpb3RzLmlu
18
334 UGFzc3dvcmQ6
18
cHVuamF1QDEyMw==
30
235 Authentication succeeded
36
MAIL FROM: <gurpartap@patriots.in>
8
250 OK

```

```

kali@kali: ~/SBT-DF203-Lab4
Version 4.9.5.0 (5.0 beta 5):
* CPU Window (still in development)
* ToDo list
* Backtrace in debugging
* Run to cursor
* Folders in Project and Class Browser
* Send custom commands to GDB
* Makefile can now be customized.
* Modified the behaviour of the -c param :=20
  -c <config file directory>
* Saving of custom syntax parameter group
* Possibility of changing compilers and tools filename.
* Many bug fixes

Version 4.9.4.1 (5.0 beta 4.1):
* back to gcc 2.95.3
* Profiling support
* new update/packages checker (vUpdate)
* Lots of bugfixes

-----=_NextPart_000_00
29
04_01CA45B0.095693F0--

28
250 OK id=1Mugho-0003Dg-Un
6
QUIT
48
221 xc90.websitewelcome.com closing connection

From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deot2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$ain>
MIME-Version: 1.0
Content-Type: multipart/mixed;
X-Mailer: Microsoft Office Outlook 12.0
Content-Type: multipart/alternative;
Content-Type: text/plain;
Content-Type: text/html;
Content-Type: text/plain;

```

```
# Replace STREAM with the relevant stream number
tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,STREAM \
| tee reports/smtp_stream_STREAM.txt

grep -Ei '^(Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|User-Agent|X-Mailer):' \
reports/smtp_stream_STREAM.txt | tee reports/message_headers.txt
```

Reconstructed payload.

```
tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,0 \
| tee reports/smtp_stream_0.txt

grep -Ei '^(Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|User-Agent|X-Mailer):' \
reports/smtp_stream_0.txt | tee reports/message_headers.txt
```

12. Part E - Determine Client, Hosts and Network Metadata

```
tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
-e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e tcp.srcport \
-e ip.dst -e tcp.dstport -e tcp.stream \
| tee reports/smtp_network_metadata.tsv

tshark -r working/smtp_working.pcap -Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
-T fields -e frame.number -e tcp.stream -e smtp.req.parameter -e data-text-lines \
| tee reports/client_indicators.tsv
```

```

kali@kali: ~/SBT-DF203-Lab4
$ tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
  -e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e tcp.srcport \
  -e ip.dst -e tcp.dstport -e tcp.stream \
  | tee reports/smtp_network_metadata.tsv

tshark -r working/smtp_working.pcap -Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
  -T fields -e frame.number -e tcp.stream -e smtp.req.parameter -e data-text-lines \
  | tee reports/client_indicators.tsv

Warning: program compiled against libxml 215 using older 214
6 1254722768.219663000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
7 1254722768.224809000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 10.10.1.4 1470 74.53.140.153 25 0
9 1254722768.566183000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
10 1254722768.568729000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
11 1254722768.911081000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
12 1254722768.911655000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
13 1254722769.253544000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
14 1254722769.254118000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
15 1254722769.613798000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
16 1254722769.614414000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
17 1254722769.959176000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
18 1254722769.957250000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
19 1254722770.319708000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
20 1254722770.320283000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
21 1254722770.661679000 00:e0:1c:3c:17:c2 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
22 1254722770.692743000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
23 1254722770.692786000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
24 1254722770.692804000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
25 1254722770.692823000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
26 1254722770.695111000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
28 1254722770.695623000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
29 1254722770.696248000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
30 1254722770.696234000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
32 1254722771.104972000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
33 1254722771.104998000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
35 1254722771.469759000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
36 1254722771.469814000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
38 1254722771.494181000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
39 1254722771.494198000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
41 1254722771.834628000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
42 1254722771.834655000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
44 1254722771.858316000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
45 1254722771.858334000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
52 1254722772.248789000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
54 1254722774.763825000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
56 1254722775.105467000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0

Warning: program compiled against libxml 215 using older 214
22 0 Line-based text data (67 lines)
26 0 Line-based text data (12 lines)

kali@kali: ~/SBT-DF203-Lab4

```

13. Part F - Encryption and Evidential Limitations

- Check whether STARTTLS is requested and whether subsequent payload becomes TLS.
- If traffic is encrypted, document metadata that remains visible: endpoints, ports, timing, certificate/session information and byte counts.
- Do not claim message content or credentials when the capture does not reveal them.
- Explain that port number alone does not prove encryption; protocol negotiation and packet dissection must be examined.

14. Required Findings Worksheet

Question	Finding
When did the SMTP session start and end?	STARTS TIME: 2026-09-14 04:09:12 (978 KB/s) - 'evidence/smtp.pcap' saved [27850/27850] END TIME: 56t2009-10-05T02:06:15.105467000-0400t74.53.140.153t10.10.1.4ttt221txc90.websi\tewelcome.com closing connection
Client IP/MAC and port	inet 172.17.0.1 /02:42:87:db:db:1d/
Server IP/MAC and port	
SMTP server banner	ESMTP Exim 4.69 #1 Mon, 05 Oc\t 2009 01:05:54
Client software	kali
Authentication method	Kali command line
Envelope sender and recipient	Subject: SMTP

Question	Finding
Message From/To/Subject	From: "Gurpartap Singh" <gurpartap@patriots.in> To: <raj_deol2002in@yahoo.co.in>
Message body type	Content-Type: multipart/mixed;
Attachment present?	
STARTTLS/TLS observed?	<pre> Relative Duration Frames Bytes Frames Bytes Frames Bytes Start 10.10.1.4:1470 <-> 74.53.140.153:25 25 1,980 bytes 28 22 kB 53 24 kB 0.036986000 7.5777 ===== </pre>
Key limitations	

Required Screenshots Checklist

- ☐ Capture hashes and capinfos summary.
- ☐ SMTP conversation list.
- ☐ 220 service-ready response.
- ☐ EHLO/HELO and AUTH exchange.
- ☐ Masked Base64 decoding result.
- ☐ MAIL FROM, RCPT TO and DATA.
- ☐ Follow TCP Stream reconstruction.
- ☐ Email headers and client software.
- ☐ Source/destination IP, ports and MAC addresses.
- ☐ TLS/STARTTLS assessment.

Student Report Template

6. Executive summary.
7. Evidence integrity and scope.
8. SMTP timeline.
9. Command/response analysis.
10. Authentication evidence and redaction method.
11. Email reconstruction.
12. Client/server and network metadata.
13. Encryption assessment and limitations.
14. Conclusions and appendix.

Submission Requirements

- One PDF report named SBT-DF203-Lab4_RegNo_FullName.pdf.
- The original or generated PCAP/PCAPNG evidence file and its SHA-256 hash text file.
- A reports folder containing exported CSV/TXT command output and any recovered objects.
- Screenshots must be readable, numbered and referenced in the report narrative.
- Compress the report and evidence outputs into one ZIP named with your registration number and lab number.
- Do not submit passwords or decoded credentials in public discussion channels; include them only in the protected assessment submission where required.

Marking Rubric (100 Marks)

Assessment Area	Marks	What the Instructor Will Check
Evidence handling	10	Original preserved, working copy and hashes documented.

Assessment Area	Marks	What the Instructor Will Check
SMTP command analysis	20	Correct commands, codes, direction and timeline.
Authentication analysis	15	Correct offline decoding and safe redaction.
Message reconstruction	25	Accurate headers, body type, client and content findings.
Network metadata and encryption	20	Correct endpoints, ports, MACs and TLS assessment.
Report quality	10	Professional, concise and evidence-linked report.

Instructor Quick Answer Guide

Checkpoint	Expected Observation
Service-ready code	220.
Client greeting	HELO or EHLO.
Authentication evidence	Often AUTH LOGIN with Base64-encoded fields in plaintext SMTP.
Message data	Begins after DATA and ends with SMTP termination sequence.
Client indicator	Slide capture identifies Microsoft Outlook 12.0.
Important limitation	Base64 is reversible encoding; TLS would hide application content.

Command Reference Summary

Command	Purpose
smtp	Wireshark display filter for SMTP.
smtp.req / smtp.rsp	Filter SMTP requests and responses.
Follow TCP Stream	Reconstruct the application conversation.
base64.b64decode	Decode Base64 offline.
capinfos file	Summarize capture metadata.
eth.addr / ip.addr / tcp.port	Endpoint filters.

Academic Integrity and Authorization Statement

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.